

Amazon Support Agent

Hiver SDE Intern Assignment — Take-Home Report
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1. Problem Framing

What “good” means for Amazon support:

- Correct intent classification so the right response is triggered
- Empathetic, actionable replies resolving issues in under 280 characters
- Safe escalation decisions — never auto-reply to legal, fraud, or payment threats

What I chose NOT to build:

- Multi-turn conversation handling (single-turn only)
- Fine-tuned classifier (LLM prompt sufficient for proof-of-concept)
- Real-time Twitter API integration
- Non-English message handling (filtered to English only)

Dataset:

- Source: Customer Support on Twitter (Kaggle) — 2.8M tweets
- Brand: AmazonHelp — #1 by volume with 169,840 responses
- Filtered to: 168,814 conversation pairs
- English-only working set: 4,697 conversations

2. System Architecture

4-stage pipeline:

Stage	Component	Method
1	Intent Classifier	LLM prompt using openai/gpt-oss-20b via Groq API
2	Retrieval (RAG)	Keyword search over 4,697 real conversations
3	Reply Drafter	LLM prompt grounded in top-3 retrieved examples
4	Escalation Decider	Hard rules plus LLM hybrid

7 Intents Defined:

Intent	Description
ORDER_STATUS	Tracking, delivery status, shipping updates
DELIVERY_ISSUE	Wrong address, not delivered, lost package

Intent	Description
RETURN_REFUND	Returns, refunds, money back requests
PAYMENT_ISSUE	Charges, billing errors, unauthorized payments
TECHNICAL_ISSUE	App, device, website problems
PRODUCT_ISSUE	Defective, wrong, or missing items received
CUSTOMER_COMPLAINT	Dissatisfaction, legal threats, escalations

3. Golden Evaluation Set

Total: 189 examples across 4 sampling strata, designed to ensure diversity and edge-case coverage.

Stratum	Count	Rationale
Random sample	80	Representative baseline
Short messages (<100 chars)	30	Edge case: vague messages
Angry messages (keyword-detected)	30	Edge case: emotional language
Order-number messages	29	Edge case: specific references
Question messages	20	Edge case: explicit queries

Labeling process:

- Auto-labeled with openai/gpt-oss-120b via Groq API
- First 52 rows manually verified by a human reviewer
- Human–LLM agreement on verified subset: approximately 85%
- Remaining rows use auto-label as ground truth (noted as a limitation)

Intent Distribution in Golden Set:

Intent	Count	Percentage
CUSTOMER_COMPLAINT	54	28.6%
DELIVERY_ISSUE	36	19.0%
ORDER_STATUS	36	19.0%
TECHNICAL_ISSUE	22	11.6%

Intent	Count	Percentage
RETURN_REFUND	15	7.9%
PAYMENT_ISSUE	15	7.9%
PRODUCT_ISSUE	11	5.8%

4. Results

4.1 Intent Classification Accuracy

System	Accuracy	Weighted F1
Trivial Baseline (always CUSTOMER_COMPLAINT)	28.6%	0.10
Keyword Baseline (regex matching)	42.3%	0.36
Our Agent (LLM)	67.7%	0.68

Improvement over trivial baseline: **+39.1%**. Improvement over keyword baseline: **+25.4%**.

4.2 Per-Intent Agent Accuracy

Intent	Accuracy	Examples
PAYMENT_ISSUE	86.7%	15
RETURN_REFUND	86.7%	15
DELIVERY_ISSUE	80.6%	36
CUSTOMER_COMPLAINT	63.0%	54
TECHNICAL_ISSUE	59.1%	22
ORDER_STATUS	55.6%	36
PRODUCT_ISSUE	54.5%	11

4.3 Reply Quality — LLM Judge (29 examples, scale 1–5)

Dimension	Score
Relevance	4.14 / 5
Tone	4.28 / 5

Dimension	Score
Actionability	4.17 / 5
Accuracy	4.76 / 5
Conciseness	4.86 / 5
Overall	4.44 / 5

Agent replies were rated better than actual Amazon replies in **20 out of 29 cases (69%)**.

5. What Is Misleading About My Headline Number

Headline number: **67.7% intent classification accuracy**.

Point 1 — Label Contamination

Ground truth labels were generated by openai/gpt-oss-120b, the same model family as our agent (openai/gpt-oss-20b). The agent and labeler share similar biases. Only 52 of 189 examples (27%) were manually verified by a human. True accuracy on fully human-verified labels would likely be lower.

Point 2 — Class Imbalance Hides Weakness

CUSTOMER_COMPLAINT dominates at 28.6% of examples, so the trivial baseline scores 28.6% just by always predicting it. Our agent scores 63% on CUSTOMER_COMPLAINT but only 54.5% on PRODUCT_ISSUE (11 examples). The headline number obscures poor performance on minority classes.

Point 3 — Batch Classification Inflates Accuracy

We classified 10 messages per API call to avoid rate limits. This means the model sees neighbouring messages in the same batch and may use cross-example context — not valid in real production, where each message arrives independently.

Point 4 — Distribution Mismatch

Our sample was drawn from 2019 Twitter data. Real incoming messages in 2026 may have very different language patterns, slang, emoji usage, and issue types compared to what was in the training and evaluation set.

Point 5 — LLM Judge Is Not Independent

The LLM judge is from the same model family that generated the replies, and may favor outputs from similar LLMs. The 4.44/5 reply score needs more human validation — 29 judge examples is too small for statistical confidence.

6. Failure Analysis — Top 5 Failure Modes

Failure Mode 1: ORDER_STATUS vs DELIVERY_ISSUE Confusion — 9 cases

Example message: “@AmazonHelp No I contacted just delayed”

True label: ORDER_STATUS | Predicted: DELIVERY_ISSUE

Hypothesis: The word “delayed” triggers delivery associations, but the customer is asking for a status update, not reporting a delivery failure. The boundary between these two intents is genuinely ambiguous in short Twitter messages.

Failure Mode 2: CUSTOMER_COMPLAINT vs TECHNICAL_ISSUE Confusion — 8 cases

Example message: “None of your links are taking me to a solution. Been after this since 3 days. Most pathetic service”

True label: CUSTOMER_COMPLAINT | Predicted: TECHNICAL_ISSUE

Hypothesis: Messages containing technical words like “links” and “solution” but expressing frustration get misclassified. The model focuses on surface keywords rather than the emotional intent of the message.

Failure Mode 3: Short and Vague Messages — 17 cases, 27.9% of all failures

Example messages: “@AmazonHelp ??” and “@AmazonHelp What is ^HA?”

True label: ORDER_STATUS | Predicted: TECHNICAL_ISSUE

Hypothesis: Messages under 80 characters give the model insufficient context to classify correctly.

Emoji-only or single-word messages are essentially unclassifiable without conversation history.

Failure Mode 4: Non-English Messages Slipping Through — observed in judge set

Example message: “@AmazonHelp Oui si je viens ici...” (French)

True label: ORDER_STATUS | Predicted: CUSTOMER_COMPLAINT

Hypothesis: Our ASCII filter (threshold >85%) still lets through some non-English messages. The model attempts classification without any language detection step.

Failure Mode 5: Sarcasm and Irony — observed

Example message: “Definitely think Amazon packed this item efficiently...” (sarcastic — item arrived damaged)

True label: DELIVERY_ISSUE | Predicted: CUSTOMER_COMPLAINT

Hypothesis: Sarcastic praise is interpreted literally by the model. There is no sarcasm detection step in our pipeline.

7. Decision Log — 15 Non-Obvious Decisions

#	Decision	Rationale
1	Chose AmazonHelp over other brands	Largest dataset (169,840 responses), most diverse intents, clearest escalation scenarios
2	Used openai/gpt-oss-20b, not 120b	The 120b model hit rate limits constantly on the free Groq tier. The 20b model has higher TPM limits with comparable classification quality
3	Used Groq API, not OpenAI	OpenAI was unavailable. Groq provides a free tier with capable models and the same prompt engineering applies
4	Chose 7 intents, not more	Tested with 10 intents and accuracy dropped significantly. 7 intents covers approximately 95% of cases cleanly with minimal overlap
5	Filtered to English only using ASCII ratio	Simple heuristic with ASCII ratio greater than 85% threshold. Removes approximately 6% of conversations effectively
6	Keyword retrieval, not embedding retrieval	Faster to build, zero additional API cost, interpretable results. Clear limitation acknowledged
7	Batch classification with 10 messages per call	Solves the rate limit problem on the free tier. Trade-off is slight context leakage between examples in the batch

#	Decision	Rationale
8	PAYMENT_ISSUE always escalated	Financial disputes require account verification. A wrong auto-reply could cause significant real harm to customers
9	Hard rules plus LLM hybrid for escalation	Pure rules miss nuanced cases. Pure LLM is expensive. The hybrid approach is fast, cheap, and auditable
10	280-character reply enforced in prompt, not post-processing	Enforced via prompt instruction to preserve natural phrasing rather than truncating
11	Confidence below 0.4 triggers escalation	Uncertain classification is itself a risk signal. Better to escalate than confidently mishandle
12	Four sampling strata for the golden set	Random alone overrepresents the majority class. Strata ensure short, angry, and order-number edge cases are covered
13	Auto-labeled, then manually verified 52 rows	Full manual labeling of 189 examples takes over 3 hours. Pragmatic trade-off explicitly noted
14	Saved progress every 10 rows during labeling	Prevents losing work on long-running jobs if a rate limit interrupts mid-run
15	Three-second sleep between judge API calls	Prevents rate limits during LLM judge evaluation. Slower but reliable on the free Groq tier

8. What I Would Do With One More Week

- Semantic retrieval: replace keyword search with text-embedding-3-small embeddings for proper similarity-based RAG retrieval
- Full human verification: verify all 189 golden examples manually for trustworthy ground truth labels
- Fine-tune DistilBERT: train a small classifier on the golden set for cheaper, faster, fully offline intent classification
- Multi-turn handling: chain conversation threads for context-aware replies using full conversation history
- Sarcasm detection: add a pre-processing step before classification to detect and handle ironic messages
- Language detection: build a separate pipeline for non-English messages instead of filtering them out
- A/B test escalation thresholds: test different confidence cutoffs against human escalation decisions on edge cases
- Expand judge validation: score 50 or more examples manually for statistically meaningful human-LLM correlation measurement